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Does Science Advance One Funeral at a Time?

Contents

1 Big picture

- **Question.** Does the death of an eminent scientist open space for outsiders to enter and redirect a scientific field, or does scientific progress mainly continue along the same intellectual path?
- **Core idea.** The paper treats the premature death of a superstar scientist as a shock to the leadership structure of narrowly defined research subfields. It asks how publication activity, entrant identity, citation impact, and research direction change after the star dies.
- **Setting.** The authors study 452 eminent life scientists who died while still actively engaged in research. They construct subfields around the scientists' own recent papers using PubMed's related-articles algorithm.
- **Main empirical design.** Each treated subfield is matched to similar control subfields whose associated superstar scientists are still alive. The core estimates compare changes in treated and control subfields after the death year.
- **Main result.** Collaborators of the deceased scientist reduce their publication activity in the affected subfields. Non-collaborators increase their publication activity, so the total field does not simply collapse.
- **Magnitude.** In the baseline Table 3 specification, the coefficient for non-collaborator publications is 0.082, implying an increase of about $100 \times (\exp(0.082) - 1) = 8.55\%$.
- **Impact result.** The rise in outsider output is disproportionately concentrated in highly cited papers. The coefficient rises monotonically across citation quantiles, reaching 0.320 for papers above the ninety-ninth percentile of vintage-specific long-run citations.
- **Direction result.** Entrants do not simply move into marginal or peripheral topics. They publish on topics close to the subfield core, but they draw on different references, avoid the deceased star's bibliome, and use newer scientific inputs.
- **Identity result.** The new contributors are not mainly old competitors who were already active in the superstar's niche. They are largely intellectual outsiders with little or no prior overlap with the star's subfield.
- **Barrier result.** Entry is strongest when the deceased star's subfield was less coherent, less socially cliquish, and less controlled through editorial channels, NIH study sections, or concentrated NIH funding.
- **Overall message.** Science can advance partly through generational turnover. Superstar scientists may shape fields not only by producing knowledge but also by exerting social and intellectual influence that limits outsider entry while they are alive.

2 Incremental contribution of the paper

2.1 Contribution relative to economics of science

- The paper contributes to the economics of science by bringing a credible empirical design to a classic question: whether incumbent scientific leaders slow or redirect the arrival of new ideas.
- Earlier work often focused on productivity effects of star scientists, collaboration networks, peer effects, or knowledge spillovers. This paper asks a different question: whether the disappearance of a star changes the opportunity set for outsiders.
- The paper adds field-level dynamics to the literature. Instead of asking only how the deceased scientist's collaborators are affected, it asks how the entire intellectual neighborhood around the star changes.

Interpretation: The paper shifts the unit of analysis from the scientist to the scientific subfield. That shift is crucial because the main effect is not just a loss of collaborator output but a reallocation of intellectual opportunity toward outsiders.

2.2 Contribution relative to sociology and history of science

- The Planck quote suggests that scientific progress may require the passing of old leaders. Historical and sociological work has long debated this idea, but systematic evidence has been limited.
- The paper turns this qualitative claim into a measurable empirical object: the entry rate of non-collaborators into narrowly defined fields after the death of a star.
- It distinguishes between several channels: loss of collaborators, entry by outsiders, changes in citation impact, changes in research direction, and changes in the barriers that shape entry.

Interpretation: The incremental contribution is not simply to confirm Planck's aphorism. It shows that the mechanism is nuanced: outsiders do enter, but they build on core topics while shifting reference bases and intellectual trajectories.

2.3 Contribution relative to identification strategies

- Deaths of actively publishing scientists provide a plausibly exogenous shock to field leadership.
- The paper strengthens this design by focusing on premature deaths, using matched control subfields, controlling for field and time trends, and examining dynamic event-study patterns.
- The authors separate collaborators from non-collaborators, which allows them to distinguish the direct productivity loss from the indirect opportunity created for outsiders.

Interpretation: The empirical design does not merely show that output changes after deaths. It identifies the composition of output changes and shows that the margin of adjustment is outsider entry.

2.4 Contribution relative to measurement

- The paper uses PubMed related-article links to construct local research subfields around individual source papers.
- It classifies contributors as collaborators or non-collaborators using coauthorship history with the superstar.
- It uses publication counts, NIH grant awards, citation quantiles, reference patterns, MeSH-term vintages, author overlap, and subfield coherence measures.
- This measurement strategy makes it possible to observe not only how much science is produced but also who produces it and whether the intellectual direction changes.

Interpretation: The paper's measurement contribution is large: it converts a broad question about scientific revolutions into observable micro-level field dynamics.

2.5 Contribution relative to policy and institutions

- The paper suggests that scientific fields may experience entry barriers created by prestige, social networks, editorial influence, and funding control.
- It implies that the organization of science can affect the direction of knowledge production, not only the quantity of research.
- It raises policy questions about grant allocation, editorial gatekeeping, and how scientific institutions can preserve intellectual openness while still benefiting from expert leadership.

Interpretation: The incremental policy contribution is that star scientists can be both engines of knowledge and bottlenecks to entry. Institutions that rely heavily on star authority may unintentionally slow intellectual renewal.

3 Data and measurement

3.1 Superstar scientists

- The study begins with eminent life scientists who died while still scientifically active.
- The final sample contains 452 deceased superstar scientists.
- A superstar is defined using indicators of scientific prominence such as prizes, high publication output, high citations, NIH funding, and membership in elite scientific communities.
- The median superstar was born in 1930, earned a degree around 1957, died around 1992, and died at age 61.
- The average superstar had about 138 career publications, 8,341 career citations, and 6.8 associated subfields.

Interpretation: The sample focuses on deaths that plausibly remove field leadership rather than ordinary retirements or low-visibility departures.

3.2 Construction of subfields

- The authors define subfields around the star's recent papers using PubMed's Related Articles function.
- A source paper by the star generates a set of intellectually proximate articles.
- These related articles define the local research subfield.
- Subfields are intentionally narrow: they are closer to scientific neighborhoods than to broad disciplines.
- The approach allows the boundaries of a subfield to adapt to the content of research rather than to administrative fields.

Interpretation: The subfield definition is central because the paper studies local intellectual leadership. A broad field such as oncology would be too coarse; a PMRA subfield isolates the neighborhood where the star could plausibly matter.

3.3 Treated and control subfields

- Treated subfields are subfields associated with deceased stars.
- Control subfields are comparable subfields associated with superstar scientists who remain alive at the relevant counterfactual date.
- The matching procedure uses baseline activity, article stock, author counts, source-article citations, source-article age, star characteristics, and field characteristics.
- Table 2 shows that treated and control subfields are closely balanced at baseline.

Interpretation: The control group is meant to approximate how treated subfields would have evolved had the star not died.

3.4 Collaborators and non-collaborators

- A collaborator is an author who previously coauthored with the star.
- A non-collaborator is an author publishing in the subfield with no such prior coauthorship link.
- Outcomes are decomposed into publications and NIH funding by all authors, collaborators, and non-collaborators.

Interpretation: This decomposition lets the paper separate a direct loss of the star's network from a broader opening of the field to outsiders.

3.5 Outcome variables

- **Publication flows:** yearly number of related articles in the subfield.
- **NIH funding flows:** number of NIH grant awards linked to the subfield.
- **Scientific impact:** citation-quantile bins of publications by non-collaborators.
- **Research direction:** intellectual proximity, references in or out of the subfield, references to the star's bibliome, and vintage of cited references or MeSH term combinations.
- **Author identity:** overlap between entrant authors' prior publications and the star's subfield.
- **Entry barriers:** intellectual coherence, social cliquishness, editorial channels, NIH study-section overlap, and concentration of NIH funding.

Interpretation: The outcome set is designed to answer three separate questions: how much entry occurs, who enters, and whether the field moves in a different direction.

4 Empirical specifications

4.1 Baseline difference-in-differences Poisson specification

The main empirical model is a conditional fixed-effects Poisson specification of the form

$$E[Y_{f,t} \mid X_{f,t}] = \exp\left(\alpha_f + \lambda_t + \gamma Age_{f,t} + \delta Post_{f,t} + \beta Treated_f \times Post_{f,t}\right), \tag{1}$$

where

- $Y_{f,t}$ is a publication or NIH funding flow in subfield f and year t;
- α_f are subfield fixed effects;
- λ_t are year effects;
- $Age_{f,t}$ captures subfield age effects;
- $Post_{f,t}$ switches on after the death year or counterfactual death year;
- $Treated_f \times Post_{f,t}$ is the treatment effect of superstar death.

Interpretation: The coefficient β is interpreted as a semi-elasticity. A value of 0.082 implies approximately 8.55% higher publication flow, because $100 \times (\exp(0.082) - 1) = 8.55\%$.

4.2 Collaborator and non-collaborator specifications

The same specification is estimated separately for:

$$Y_{f,t}^{All}, \qquad Y_{f,t}^{Collaborators}, \qquad Y_{f,t}^{Noncollaborators}. \tag{2}$$

Interpretation: This is the key decomposition. If death merely destroys scientific capacity, collaborator and non-collaborator flows should both decline. If death opens a field, collaborators should decline while non-collaborators increase.

4.3 Dynamic event-study specification

The dynamic version replaces the single treatment effect with relative-year indicators:

$$E[Y_{f,t} \mid X_{f,t}] = \exp\left(\alpha_f + \lambda_t + \gamma Age_{f,t} + \sum_{k=-K}^{K} \beta_k Treated_f \times 1\{t - T_f = k\}\right). \tag{3}$$

Interpretation: The event-study coefficients test for pre-trends and show whether effects are immediate, delayed, temporary, or persistent.

4.4 Citation impact specifications

For impact tests, the dependent variable counts non-collaborator publications in citation bins:

$$Y^q_{f,t} = \#\{\text{non-collaborator papers in subfield } f \text{ and year } t \text{ in citation quantile } q\}. \tag{4}$$

Interpretation: This asks whether entry after a star's death is low-quality marginal entry or high-impact entry. The paper finds that the effect is strongest in high-citation bins.

4.5 Research direction specifications

The paper partitions non-collaborator publications by several content measures:

- PMRA-proximate versus PMRA-distant articles;
- articles citing in-field versus out-of-field references;
- articles citing the star's bibliome versus not citing it;
- articles using young versus old references or MeSH-term combinations.

Interpretation: These specifications ask whether outsiders simply add more papers or whether they change the knowledge base and intellectual direction of the subfield.

4.6 Entry barrier specifications

The sample is split by baseline measures of barriers:

- intellectual coherence measured by PMRA or citation structure;
- social cliquishness;
- editorial channels;
- NIH study-section channels;
- concentration of NIH funding.

Interpretation: The barrier specifications ask whether stars block entry directly through intellectual dominance or indirectly through social, editorial, and funding networks.

5 Identification and empirical design

5.1 Identifying assumption

The identifying assumption is that, absent the star's death, treated subfields would have followed trends similar to matched control subfields.

Interpretation: The matched controls and pre-trend event-study evidence support this assumption, although no observational design can completely rule out all field-specific shocks.

5.2 Why death is useful for identification

- Death removes a scientist unexpectedly relative to the normal career cycle.
- It is not chosen by the scientist, collaborators, or entrants.
- The paper focuses on deaths while scientists are active, reducing the concern that the star had already exited the field.
- The design compares treated subfields to control subfields linked to living stars.

Interpretation: Death provides a rare event that changes leadership without being selected by the research field itself.

5.3 Main threats

- Some deaths may be anticipated, giving collaborators time to adjust or heirs to inherit the field.
- Stars may be associated with declining or maturing fields.
- Subfields may be defined partly by the star's own output, which could mechanically change after death.
- Control subfields may differ in unobserved ways despite matching.

Interpretation: The paper addresses these threats with control subfields, event studies, cause-of-death robustness checks, exclusion of the star's own papers, and star-level robustness analyses.

5.4 What the design can and cannot prove

- It can show that outsider entry rises after star death relative to matched controls.
- It can show that the rise is concentrated in high-impact and intellectually fresh work.
- It can show that stronger barriers predict less entry.
- It cannot directly observe whether the star intentionally blocked entry.
- It cannot prove that every entrant was deterred by the star while alive; it identifies a field-level pattern consistent with the barrier interpretation.

Interpretation: The paper's strength lies in combining multiple margins of evidence: quantity, impact, direction, entrant identity, and barriers.

6 Tables and figures

6.1 Table 1 and Table 2: sample construction and matched baseline

Read:

- Table 1 describes the 452 deceased superstar scientists.
- The median superstar dies at age 61 and is associated with four subfields.
- The average superstar has 138 career publications, 8,341 career citations, and over $16 million in NIH funding.
- Table 2 compares control and treated subfields at baseline.
- Treated subfields include 3,076 subfields; control subfields include 31,142 observations.
- Treated and control subfields are very similar in baseline article stock, author counts, source-article citations, and star characteristics.

Economic message: The empirical comparison is credible only if treated and control subfields are similar before death. Tables 1 and 2 show the sample is composed of prominent scientists and that the matching procedure creates reasonable baseline balance.

Table 1—Summary Statistics: Deceased Superstar Scientists

	Mean	Median	SD	Min.	Max.
Year of birth	1930.157	1930	11.011	1899	1959
Degree year	1957.633	1957	11.426	1928	1986
Year of death	1991.128	1992	8.055	1975	2003
Age at death	60.971	61	9.778	34	91
Female	0.102	0	0.303	0	1
MD degree	0.403	0	0.491	0	1
PhD degree	0.489	0	0.500	0	1
MD/PhD degree	0.108	0	0.311	0	1
Sudden death	0.409	0	0.492	0	1
Number of subfields	6.805	4	7.308	1	57
Career number of pubs.	138.221	112	115.704	12	1,380
Career number of citations	8,341	5,907	8,562	120	72,122
Career NIH funding	$16,637,919	$10,899,139	$25,441,933	0	$329,968,960
Sits on NIH study section	0.007	0	0.081	0	1
Career number of editorials	0.131	0	0.996	0	17

Notes: Sample consists of 452 superstar life scientists who died while still actively engaged in research. See online Appendix A for more details on sample construction.

Table 1: deceased superstar scientists

Table 2—Summary Statistics: Control and Treated Subfields at Baseline

	Mean	Median	SD	Min.	Max.
Control subfields (observations = 31,142)					
Baseline stock of related articles in the field	76.995	59	64.714	0	384
Baseline stock of related articles in the field, non-collaborators	68.390	51	60.222	0	381
Baseline stock of related articles in the field, collaborators	8.604	5	10.358	0	125
Source article number of authors	3.970	4	1.901	1	15
Source article citations at baseline	16.331	8	30.305	0	770
Source article long-run citations	70.427	38	116.108	1	4,495
Investigator gender	0.067	0	0.249	0	1
Investigator year of degree	1960.546	1962	10.998	1926	1991
Death year	1991.125	1991	7.968	1975	2003
Age at death	58.100	58	8.795	34	91
Investigator cumulative number of publications	164	131	123	1	1,109
Investigator cumulative NIH funding at baseline	$18,784,517	$11,904,846	$25,160,518	0	$387,558,656
Investigator cumulative number of citations	12,141	8,010	12,938	9	157,581
Treated subfields (observations = 3,076)					
Baseline stock of related articles in the field	76.284	58	64.046	0	368
Baseline stock of related articles in the field, non-collaborators	67.752	51	59.725	0	357
Baseline stock of related articles in the field, collaborators	8.532	5	9.841	0	86
Source article number of authors	3.987	4	1.907	1	14
Source article citations at baseline	16.694	8	36.334	0	920
Source article long-run citations	70.432	35	180.528	1	6,598
Investigator gender	0.099	0	0.299	0	1
Investigator year of degree	1960.141	1961	10.898	1928	1986
Death year	1991.125	1991	7.970	1975	2003
Age at death	58.100	58	8.796	34	91
Investigator cumulative number of publications	170	143	118	12	1,380
Investigator cumulative NIH funding at baseline	$17,637,726	$12,049,690	$24,873,018	0	$329,968,960
Investigator cumulative number of citations	11,580	8,726	10,212	120	72,122

Notes: The sample consists of subfields for 452 deceased superstar life scientists and their matched control subfields. See online Appendix D for details on the matching procedure. All time-varying covariates are measured in the year of superstar death.

Table 2: matched control and treated subfields

6.2 Figure 1 and Table 3: baseline treatment effect

Read:

- Figure 1 shows that treated and control subfields have similar distributions of publication stock before the death event.
- Table 3 shows that total publication flows rise only modestly after death.
- Collaborator publication flows decline sharply: coefficient -0.232.
- Non-collaborator publication flows rise: coefficient 0.082, or approximately 8.55 percent.

- NIH grant flows show a similar pattern: collaborators decline and non-collaborators increase.

Economic message: The death of a superstar does not simply shrink the field. It reallocates activity away from collaborators and toward outsiders.

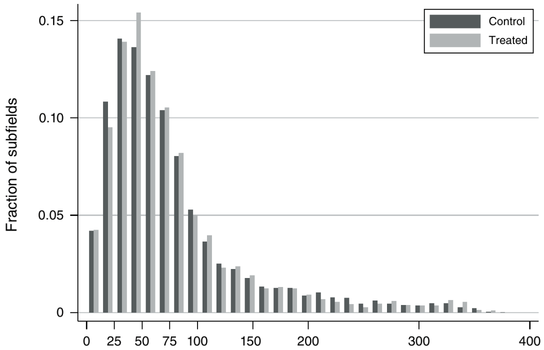


FIGURE 1. CUMULATIVE STOCK OF PUBLICATIONS AT TIME OF DEATH

: We compute the cumulative number of publications, up to the year that immediately precedes the
(or counterfactual year of death) for the 3,076 treated subfields and 31,142 control subfields.

Figure 1: cumulative publication stock

TABLE 3—EFFECT OF SUPERSTAR DEATH ON SUBFIELD ENTRY RATES

	Publication flows			NIH funding flows (number of awards)		
	All authors	Collaborators only	Non-collaborators only	All authors	Collaborators only	Non-collaborators only
	(1)	(2)	(3)	(4)	(5)	(6)
After death	0.051	−0.232	0.082	0.046	−0.265	0.110
	(0.029)	(0.057)	(0.029)	(0.035)	(0.076)	(0.033)
Number of investigators	6,260	6,124	6,260	6,215	5,678	6,202
Number of fields	34,218	33,096	34,218	33,912	29,163	33,806
Number of field-year obs.	1,259,176	1,217,905	1,259,176	1,049,942	902,873	1,046,678
log likelihood	−2,891,110	−729,521	−2,768,252	−1,350,204	−472,329	−1,223,913

Notes: Estimates stem from conditional (subfield) fixed effects Poisson specifications. The dependent variable is the total number of publications in a subfield in a particular year (columns 1, 2, and 3), or the total number of NIH grants that acknowledge a publication in a subfield (columns 4, 5, and 6). All models incorporate a full suite of year effects and subfield age effects, as well as a term common to both treated and control subfields that switches from 0 to 1 after the death of the star, to address the concern that age, year, and individual fixed effects may not fully account for trends in subfield entry around the time of death. Exponentiating the coefficients and differencing from one yield numbers interpretable as elasticities. For example, the estimates in column 3 imply that treated subfields see an increase in the number of contributions by non-collaborators after the superstar passes away, a statistically significant $100 \times (\exp[0.082] - 1) = 8.55\%$. Robust standard errors in parentheses, clustered at the level of the star scientist.

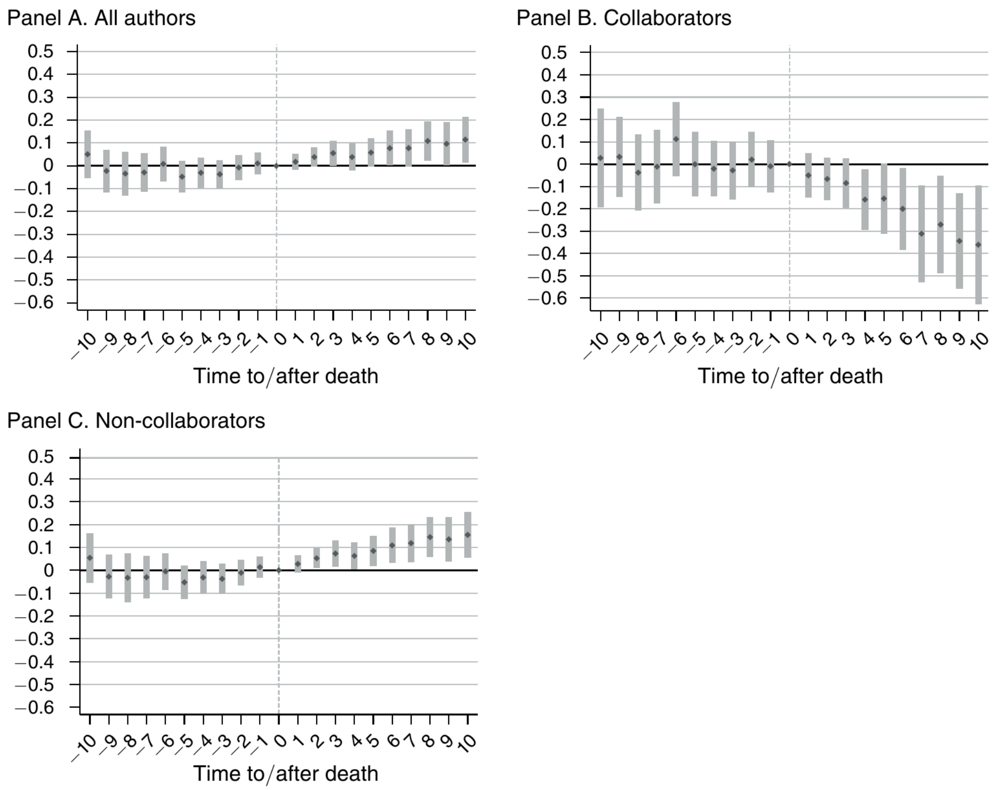
Table 3: effect on entry rates

6.3 Figure 2: dynamic pattern of entry

Read:

- Panel A shows the dynamic effect for all authors.
- Panel B shows a sustained decline in collaborator activity after the star dies.
- Panel C shows a gradual increase in non-collaborator activity after death.
- The pre-death coefficients are relatively flat, supporting the parallel-trends interpretation.

Economic message: The increase in outsider entry is not a one-year blip. It grows over time, suggesting a real opening of the subfield rather than a temporary publication spike.



6.4 Table 4: scientific impact of entrant publications

Read:

- Table 4 breaks non-collaborator publications into vintage-specific long-run citation quantiles.
- The overall non-collaborator effect is 0.082.
- The effect is negative or close to zero for low-impact publications.
- The coefficient rises to 0.125 for papers between the seventy-fifth and ninety-fifth percentiles.
- It rises further to 0.232 for papers between the ninety-fifth and ninety-ninth percentiles.
- It reaches 0.320 for papers above the ninety-ninth percentile.

Economic message: Outsider entry after superstar death is not merely low-quality opportunistic entry. The largest effects are in highly cited work.

6.5 Table 5: entry and research direction

Read:

- Panel A shows that the new non-collaborator contributions are strongest for intellectually proximate articles, not distant peripheral articles.
- Panel B shows growth in articles that cite outside the PMRA subfield and in articles that do not cite the star's bibliome.
- Panel C shows that new contributions are more likely to cite younger references and use newer MeSH-term combinations.

Table 4—Scientific Impact of Entry

	Vintage-specific long-run citation quantile						
	All pubs	Bottom quartile	2nd quartile	3rd quartile	Btw. 75th and 95th percentile	Btw. 95th and 99th percentile	Above 99th percentile
After death	0.082	−0.028	0.008	0.031	0.125	0.232	0.320
	(0.029)	(0.036)	(0.033)	(0.032)	(0.035)	(0.049)	(0.081)
Number of investigators	6,260	6,222	6,260	6,257	6,255	6,161	5,283
Number of fields	34,218	33,714	34,206	34,212	34,210	33,207	21,852
Number of field-year observations	1,259,176	1,240,802	1,258,738	1,258,954	1,258,880	1,221,952	804,122
log likelihood	−2,768,252	−689,465	−1,125,555	−1,432,223	−1,469,096	−542,735	−156,519

Notes: Estimates stem from conditional (subfield) fixed effects Poisson specifications. The dependent variable is the total number of publications by non-collaborators in a subfield in a particular year, where these publications fall in a particular quantile bin of the long-run, vintage-adjusted citation distribution for the universe of journal articles in *PubMed*. All models incorporate a full suite of year effects and subfield age effects, as well as a term common to both treated and control subfields that switches from 0 to 1 after the death of the star. Exponentiating the coefficients and differencing from one yield numbers interpretable as elasticities. For example, the estimates in column 1, panel A imply that treated subfields see an increase in the number of contributions by non-collaborators after the superstar passes away, a statistically significant $100 \times (\exp[0.082] - 1) = 8.55\%$. Robust standard errors in parentheses, clustered at the level of the star scientist.

Table 5—Entry and Research Direction

	Cardinal measure		Ordinal measure	
	Intellectual proximate articles	Intellectual distant articles	Intellectual proximate articles	Intellectual distant articles
Panel A				
After death	0.091	0.028	0.117	−0.024
	(0.030)	(0.035)	(0.028)	(0.037)
Number of investigators	6,228	6,099	6,260	6,017
Number of fields	33,375	32,232	34,218	31,712
Number of field-year observations	1,228,157	1,186,589	1,259,176	1,167,423
log likelihood	−1,628,374	−1,816,449	−1,893,982	−1,628,170
	In-field versus out-of-field references		Backward citations to the star's bibliome	
	With in-field references	Without in-field references	With references to the star	Without references to the star
Panel B				
After death	−0.023	0.128	0.078	0.152
	(0.041)	(0.031)	(0.036)	(0.034)
Number of investigators	6,195	6,260	6,247	6,259
Number of fields	32,721	34,218	34,179	34,147
Number of field-year observations	1,204,315	1,259,176	1,257,747	1,256,576
log likelihood	−792,795	−2,510,344	−1,914,448	−1,767,571
	Vintage of cited references		Vintage of 2-way MeSH term combinations	
	Young	Old	Young	Old
Panel C				
After death	0.071	−0.010	0.090	0.029
	(0.035)	(0.034)	(0.033)	(0.036)
Number of investigators	6,260	6,260	6,258	6,260
Number of fields	34,218	34,214	34,206	34,210
Number of field-year observations	1,259,176	1,259,044	1,258,732	1,258,906
log likelihood	−2,124,598	−1,613,457	−1,853,062	−1,784,275

Economic message: New entrants remain close to the core problems of the subfield, but they bring different intellectual inputs. The field is rejuvenated without completely abandoning its core subject matter.

6.6 Figure 3: who enters?

Read:

- Panel A shows that about half of related articles are written by authors with zero prior overlap with the star's subfield.
- Panel B shows that the treatment effect is largest for new scientists and low-overlap authors.
- The increase is not mainly driven by old rivals already embedded in the subfield.

Economic message: The authors driving post-death growth are largely outsiders. Star death appears to open the field to entrants who had previously been at the edge of the intellectual neighborhood.

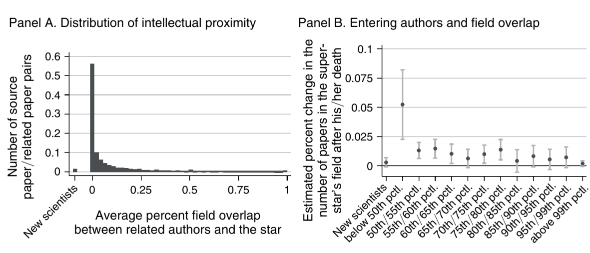


Figure 3. Characteristics of Related Authors: Competitors or Outsiders?

Notes: Panel A displays the distribution of overlap between the past output of related authors and each star's subfield. For each author on a related article matched to the AAMC Faculty Roster, we create a metric of intellectual proximity by computing the fraction of their publications that belongs to the star's subfield. Slightly more than half of related articles have authors with zero overlap, i.e., this related article is their first contribution to the star's subfield. 1.24 percent of related articles are authored by new scientists for whom this publication within the subfield is also their first publication overall. Using this information, we aggregate the number of related articles in a particular subfield and in a particular year, e.g., *the number of articles in the subfield in year t that have authors above the ninety-fifth percentile in our measure of field overlap*. In panel B, each dot corresponds to the magnitude of the treatment effect in a separate regression where the dependent variable is the number of articles in each subfield authored by scientists who belong to a particular bin of intellectual proximity, as measured by field overlap above.

Table 6—Breakdown by Star Scientist Characteristics

	Publications		Citations		Funding		Importance to the field	
	Below median	Above median	Below median	Above median	Below median	Above median	Below median	Above median
After death	0.059	0.116	0.036	0.125	0.014	0.162	0.063	0.123
	(0.037)	(0.050)	(0.042)	(0.040)	(0.040)	(0.052)	(0.031)	(0.045)
Number of investigators	2,901	4,836	2,792	4,619	3,048	4,287	5,019	4,493
Number of fields	17,210	17,008	17,328	16,890	15,731	15,487	16,985	17,233
Number of field-year observations	632,089	627,087	636,750	622,426	578,277	570,665	625,140	634,036
log likelihood	−1,377,727	−1,387,650	−1,367,335	−1,396,652	−1,268,559	−1,252,952	−1,462,538	−1,257,973

Notes: Estimates stem from conditional (subfield) fixed effects Poisson specifications. The dependent variable is the total number of publications by non-collaborators in a subfield in a particular year. Each pair of columns splits the sample across the median of a particular covariate for the sample of fields (treated and control) in the baseline year. The table examines differences in the extent to which the eminence of the star at death (respectively counterfactual year of death for controls) influences the rate at which non-collaborators enter the field after the star passes away. Eminence is measured through the star's cumulative number of publications, the star's cumulative number of citations garnered up to the year of death, and the star's cumulative amount of NIH funding. We also have a "local" measure of eminence: the star's importance to the field, which is defined as the proportion of articles in the subfield up to the year of death for which the star is an author. All models incorporate a full suite of year effects and subfield age effects, as well as a term common to both treated and control subfields that switches from 0 to 1 after the death of the star. Exponentiating the coefficients and differencing from one yield numbers interpretable as elasticities. For example, the estimate in the second column implies that treated subfields see an increase in the number of contributions by non-collaborators after the superstar passes away, a statistically significant $100 \times (\exp[0.116] - 1) = 12.30\%$. Robust standard errors in parentheses, clustered at the level of the star scientist.

Subfield Coherence.—Entry into a field, even after it has lost its star, may be deterred if the subfield appears unusually coherent to outsiders. A subfield is likely to be perceived as *intellectually* coherent, when the researchers active in it agree on the set of questions, approaches, and methodologies that propel the field forward.

6.7 Table 6: star scientist characteristics

Read:

- Table 6 splits subfields by measures of star eminence.
- For stars with above-median publications, the effect is 0.116, compared with 0.059 below median.
- For stars with above-median citations, the effect is 0.125, compared with 0.036 below median.
- For stars with above-median NIH funding, the effect is 0.162, compared with 0.014 below median.
- For stars with above-median importance to the field, the effect is 0.123, compared with 0.063 below median.

Economic message: The opening to outsiders is strongest when the deceased scientist was especially eminent. This supports the idea that star power itself can shape entry.

6.8 Table 7: nature of entry barriers

Read:

- Panel A shows stronger entry when subfields are less intellectually coherent or less cliquish.
- The PMRA-based coherence split shows an effect of 0.202 below median versus 0.067 above median.
- The citation-based coherence split shows 0.161 below median versus 0.096 above median.
- The cliquishness split shows 0.129 below median versus 0.064 above median.
- Panel B shows larger effects when the star has weaker indirect control through editorials, NIH study sections, or NIH funding dominance.

Economic message: Outsider entry is easier when intellectual, social, and resource barriers are lower. Star influence appears to operate partly through field structure and control over publication or funding channels.

TABLE 7—THE NATURE OF ENTRY BARRIERS

	PMRA-based definition		Citation-based definition		Cliquishness	
	Below median	Above median	Below median	above median	Below median	Above median
Panel A. Subfield coherence						
After death	0.202	0.067	0.161	0.096	0.129	0.064
	(0.038)	(0.048)	(0.053)	(0.041)	(0.049)	(0.052)
Number of investigators	3,353	3,203	3,422	3,157	2,865	3,561
Number of fields	9,062	7,828	8,731	8,159	8,044	8,846
Number of field-year observations	334,142	288,284	321,826	300,600	296,704	325,722
log likelihood	−711,335	−664,170	−760,842	−631,287	−692,330	−685,682
	Editorial channel		**NIH study section channel**		**Fraction of subfield NIH funding**	
	Below median	Above median	Below median	Above median	Below median	Above median
Panel B. Indirect control through collaborators						
After death	0.147	0.086	0.134	−0.078	0.174	0.084
	(0.056)	(0.048)	(0.043)	(0.095)	(0.051)	(0.051)
Number of investigators	3,452	2,068	4,385	664	3,559	2,525
Number of fields	11,110	5,780	15,338	1,552	9,863	7,027
Number of field-year observations	410,025	212,401	565,219	57,207	363,690	258,736
log likelihood	−951,705	−461,769	−1,293,997	−125,950	−840,777	−545,782

Notes: Estimates stem from conditional (subfield) fixed effects Poisson specifications. The dependent variable is the total number of publications by non-collaborators in a subfield in a particular year. The sample is limited to the subfields in which the most eminent among the stars were active (specifically, above the median of the "cumulative citations up to the year of death" metric). Each pair of columns splits the sample across the median of a particular covariate for the sample of subfields (treated and control) in the baseline year. For example, the first two columns of panel B compare the magnitude of the treatment effect for stars whose collaborators have written an above-median number of editorials in the five years preceding the superstar's death, versus a below-median number of editorials. All models incorporate a full suite of year effects and subfield age effects, as well as a term common to both treated and control subfields that switches from 0 to 1 after the death of the star. Exponentiating the coefficients and differencing from one yield numbers interpretable as elasticities. For example, the estimates in the first column of panel B imply that treated subfields see an increase in the number of contributions by non-collaborators after the super-

7 Short summary

- The paper studies whether scientific fields change after eminent scientists die.
- It uses 452 deceased superstar life scientists and constructs narrow subfields around their recent publications.
- Treated subfields are matched to similar control subfields associated with living stars.
- Collaborators reduce their activity after the star dies.
- Non-collaborators increase their publication and NIH funding activity.
- The non-collaborator increase is strongest for highly cited publications.
- Entrants work on core subfield topics but cite different and newer scientific inputs.
- New contributors are largely intellectual outsiders, not established competitors in the same niche.
- Entry is stronger when the star was more eminent but when the subfield had lower barriers to entry.
- The paper provides evidence that star scientists can slow or shape the evolution of fields while alive, and that their death can allow fields to move in new directions.

8 Conclusion

8.1 Core conclusion

Azoulay, Fons-Rosen, and Graff Zivin show that the death of a superstar scientist reshapes the intellectual dynamics of affected subfields. Collaborators withdraw, but outsiders enter. The new entrants produce work that is often highly cited and intellectually fresh.

8.2 Economic intuition

The intuition is that stars are not only knowledge producers; they are also central nodes in social, intellectual, editorial, and funding networks. Their influence can coordinate a field, but it can also make entry by outsiders difficult. When the star disappears, outsiders face less resistance and the field can move in new directions.

8.3 Incremental contribution takeaway

The paper's incremental contribution is to show that scientific progress is shaped by the organization of attention and authority. It provides systematic evidence for a Planck-like mechanism: fields sometimes advance not because incumbents are persuaded, but because their disappearance changes who can enter and what ideas can gain traction.

8.4 Broader takeaway

The paper cautions against viewing scientific progress as only a pure contest of ideas. Institutions, status, networks, and control over publication or funding channels can shape the path of knowledge. Generational turnover can therefore be an engine of intellectual renewal.